Research activity

Fluctuations of a frequency — 40 min.

A fair coin is tossed independently. For every integer $n \geq 1$, let F_n be the frequency of "heads" among the first n tosses.

1. Write a program that simulates 1000 tosses, then plot the points $(n; F_n)$ for $1 \leq n \leq 1000$. Does the deviation from $\frac{1}{2}$ necessarily decrease at every toss?
2. Express $E(F_n)$ and $V(F_n)$ in terms of n.
3. Using Bienaymé–Chebyshev, find an upper bound for $P(|F_{1000} - \frac{1}{2}| \geq 0.05)$.
4. Determine an integer n sufficient for $P(|F_n - \frac{1}{2}| < 0.05) \geq 0.99$.
5. State what the simulated trajectory shows and what the inequality obtained guarantees. Does a single simulation constitute a proof of this guarantee?